CYBER SECURITY INTERNSHIP

TASK 1: SCAN YOUR LOCAL NETWORK FOR OPEN PORTS

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Project: Network Reconnaissance and Port Scanning
Tool: Nmap
Operating System: Kali Linux
Category: Network Security
Internship Task: 01

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1. Introduction

Network reconnaissance is an important part of cybersecurity. It involves gathering information about devices, network services, and exposed ports within a network.

In this task, Nmap (Network Mapper) is used to discover active devices and identify open TCP ports on an authorized local network.

The purpose is to understand network exposure, identify potentially risky services, and learn basic network security assessment techniques.

2. Objective

The main objectives of this task are:

- Discover active devices on a local network.
- Identify open TCP ports.
- Understand the services running on discovered ports.
- Perform TCP SYN scanning using Nmap.
- Analyze potential security risks associated with open ports.
- Understand the role of firewalls in network security.
- Save and document scan results.

3. Tools and Technologies Used

3.1 Nmap

Nmap (Network Mapper) is a free and open-source network discovery and security auditing tool.

It helps security professionals:

- Discover active network hosts.
- Identify open, closed, and filtered ports.
- Detect running services.
- Identify service versions.
- Perform network reconnaissance.

Official Website: https://nmap.org/

3.2 Wireshark (Optional)

Wireshark is a network protocol analyzer used to capture and inspect network traffic.

It can help analyze TCP packets, SYN requests, SYN-ACK responses, and RST packets.

Official Website: https://www.wireshark.org/

3.3 Kali Linux

Kali Linux is a Linux distribution designed for penetration testing, digital forensics, and cybersecurity assessments.

4. Understanding Important Concepts

4.1 What is an IP Address?

An IP address is a numerical identifier assigned to a device connected to a network.

Example:

192.168.1.105

4.2 What is a Port?

A port is a logical communication endpoint used by network services.

Some common ports include:

Port	Service	Purpose
21	FTP	File Transfer
22	SSH	Secure Remote Access
23	Telnet	Remote Access
25	SMTP	Email Transmission
53	DNS	Domain Name Resolution
80	HTTP	Web Communication
443	HTTPS	Secure Web Communication
445	SMB	File Sharing
3389	RDP	Remote Desktop

4.3 What is an Open Port?

An open port indicates that a service is listening for network connections on that port.

Open ports are not automatically vulnerabilities, but unnecessary or improperly secured services may increase security risks.

4.4 What is Network Reconnaissance?

Network reconnaissance is the process of collecting information about network devices, services, and potential entry points.

Port scanning is one of its fundamental techniques.

5. Practical Implementation

Step 1: Verify Nmap Installation

First, open the Kali Linux terminal and check whether Nmap is installed.

Command:

```
None
nmap --version
```

If Nmap is not installed, use:

```
None
sudo apt update
sudo apt install nmap -y
```

Step 2: Identify the Local IP Address

Execute:

```
None
ifconfig
```

Alternatively:

```
None
ip addr
```

Identify the network interface, IPv4 address, and default gateway.

Example:

- Local IP: 192.168.1.105
- Gateway: 192.168.1.1
- Subnet Mask: 255.255.255.0
- Network Range: 192.168.1.0/24

Note: These are example values. The actual network range must be calculated using the system's real IP address and subnet mask.

Step 3: Perform TCP SYN Scan

Execute the following command against the authorized local network:

```
None
sudo nmap -sS 192.168.1.0/24
```

Explanation:

- `sudo`: Executes Nmap with elevated privileges.
- `nmap`: Starts the Network Mapper tool.
- `-sS`: Performs a TCP SYN scan.
- `192.168.1.0/24`: Specifies the target network range.

Step 4: Save Scan Results

To save the scan results in a normal text file:

```
None
sudo nmap -sS 192.168.1.0/24 -oN scan_results.txt
```

To save results in XML format:

```
None
sudo nmap -sS 192.168.1.0/24 -oX scan_results.xml
```

To save multiple output formats:

```
None
sudo nmap -sS 192.168.1.0/24 -oA local_network_scan
```

These commands should use the actual authorized network range.

6. Understanding TCP SYN Scanning

TCP SYN scanning is a commonly used Nmap scanning technique.

It is also known as a half-open scan because the complete TCP three-way handshake is not established.

The scanning process:

1. Nmap sends a SYN packet to the target port.

2. If the port is open, the target generally responds with SYN-ACK.
3. Nmap sends an RST packet instead of completing the handshake.
4. If the port is closed, the target generally responds with RST.

TCP Three-Way Handshake

- SYN: Client requests a connection.
- SYN-ACK: Server acknowledges the request.
- ACK: Client completes the connection.

7. TCP vs UDP Scanning

TCP Scanning	UDP Scanning
Uses TCP protocol	Uses UDP protocol
Connection-oriented	Connectionless
SYN scanning is commonly used	UDP scanning uses -sU
Generally easier to interpret	Can be difficult to interpret
Uses TCP flags	Does not use TCP flags

Example commands:

TCP SYN Scan:

```
None
sudo nmap -sS TARGET
```

UDP Scan:

```
None
sudo nmap -sU TARGET
```

8. Analysis of Common Open Ports

Port	Service	Potential Security Risk

21	FTP	Credentials may be transmitted without encryption
22	SSH	Weak authentication or exposed remote access
23	Telnet	Unencrypted communication
80	HTTP	Unencrypted web traffic
445	SMB	Potential file-sharing exposure and lateral movement
3389	RDP	Unauthorized remote desktop access

The presence of an open port does not automatically indicate a security vulnerability. Risk depends on the service configuration, authentication, software version, network exposure, and access restrictions.

9. Security Recommendations

The following security measures can help reduce network exposure:

1. Disable unnecessary network services.
2. Close unused ports.
3. Configure firewall rules.
4. Use strong passwords and multi-factor authentication where supported.
5. Regularly update operating systems and network services.
6. Restrict remote access to trusted IP addresses.
7. Replace insecure protocols such as Telnet and FTP with secure alternatives.
8. Monitor network activity for suspicious connections.
9. Regularly conduct authorized network security assessments.

10. Role of a Firewall

A firewall monitors and controls incoming and outgoing network traffic according to predefined security rules.

It can:

- Allow legitimate network communication.
- Block unauthorized connections.
- Restrict access to specific ports.
- Filter traffic based on IP addresses and protocols.
- Reduce exposure to network-based attacks.

11. Wireshark Packet Analysis (Optional)

Wireshark can complement Nmap by providing visibility into network packets.

Useful display filters:

Identify TCP SYN packets:

```
None
tcp.flags.syn == 1
```

Identify TCP SYN-ACK packets:

```
None
tcp.flags.syn == 1 && tcp.flags.ack == 1
```

Identify TCP RST packets:

```
None
tcp.flags.reset == 1
```

Wireshark helps understand how network devices communicate during port scanning.

12. Practical Scan Results

This section must be completed using actual scan output.

- Local IPv4 Address: [Insert Actual IP]
- Network Range: [Insert Actual Network Range]
- Number of Active Devices: [Insert Actual Count]
- Open Ports: [Insert Actual Ports]
- Identified Services: [Insert Actual Services]

Scan Observations

[Describe the devices, ports, and services discovered during the scan.]

Security Findings

[Document any unnecessary exposed services or potential security concerns. Do not classify a port as vulnerable without supporting evidence.]

13. Interview Questions and Answers

Q1. What is an open port?

An open port is a network communication endpoint where a service is actively listening and accepting connections.

Q2. How does Nmap perform a TCP SYN scan?

Nmap sends a SYN packet to the target port. An open port generally responds with SYN-ACK, while a closed port generally responds with RST. Nmap does not complete the full TCP handshake.

Q3. What risks are associated with open ports?

Open ports may expose services to unauthorized access, exploitation, malware attacks, and data breaches if they are improperly configured or vulnerable.

Q4. Explain the difference between TCP and UDP scanning.

TCP scanning examines TCP services and connection states, while UDP scanning identifies UDP services without using a TCP handshake.

Q5. How can open ports be secured?

Open ports can be secured through firewall rules, strong authentication, software updates, service hardening, and restricting unnecessary network access.

Q6. What is a firewall's role regarding ports?

A firewall controls network traffic and can allow or block connections based on ports, protocols, and IP addresses.

Q7. What is port scanning and why do attackers perform it?

Port scanning is a reconnaissance technique used to identify open ports and running services. Attackers use it to discover potential entry points, while security professionals use it for authorized assessments.

Q8. How does Wireshark complement port scanning?

Nmap identifies open ports and services, whereas Wireshark captures and analyzes network packets to provide deeper insight into network communication.

14. Conclusion

This task provided practical experience with network reconnaissance and port scanning using Nmap.

The exercise demonstrated how to identify active devices, discover open TCP ports, understand running services, and analyze potential security risks.

It also provided an understanding of TCP SYN scanning, TCP and UDP protocols, firewall protection, and network packet analysis using Wireshark.

These skills form an important foundation for penetration testing, vulnerability assessment, and network security monitoring.

15. References

- Nmap Official Documentation: https://nmap.org/book/man.html
- Nmap Download: https://nmap.org/download.html
- Wireshark Official Website: https://www.wireshark.org/
- Kali Linux Official Website: https://www.kali.org/
- TCP/IP Networking Concepts: https://www.rfc-editor.org/
